

Conditional Fault-Tolerance of Hamiltonian Cycles for Crossed Cubes

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Abstract—¹

The crossed cube, which is a variation of the hypercube, possesses some properties superior to the hypercube. In this paper, we have prove that the graph $CQ_n - F_v - F_e$ is Hamiltonian where the degree of every fault-free vertex is at least 2 for $F_v \subset V(CQ_n)$, $F_e \subset E(CQ_n)$, $|F_v| + |F_e| \leq 2n - 6$ and $n \geq 5$.

Index Terms—Conditional fault; Crossed cube; Hamiltonian cycle embedding; Faulty nodes; Faulty edges; Hypercube; Interconnection network

I. INTRODUCTION

The crossed cube is the superior graph which transfers from the hypercube graph because it has some appealing features such as shorter diameter, the crossed cube network has received considerable research attention [1–14].

The crossed cube is a Hamiltonian graph. There are many research results for the fault-tolerance of crossed cubes. Let CQ_n denote the n -dimensional crossed cube and F_v the set of faulty vertices of CQ_n and F_e the set of faulty edges of CQ_n . In [7], the authors proved that there exists a fault-free Hamiltonian cycle in a $CQ_n - F_v - F_e$ with $|F_v| + |F_e| \leq n - 2$, and there exists a fault-free Hamiltonian path between any pair of nodes in a $CQ_n - F_v - F_e$ with $|F_v| + |F_e| \leq n - 3$ for $n \geq 3$.

When $|F_v| + |F_e| = n - 1$, in fact, the graph $CQ_n - F_v - F_e$ is Hamiltonian if the degree of every fault-

free vertex is at least 2. Some researchers studied the conditional fault-tolerance for crossed cubes. In [8], the authors showed that $CQ_n - F_e$ is Hamiltonian for $|F_e| \leq 2n - 5$ if the degree of every fault-free vertex is at least 2. Dong and Yang showed that there exists a fault-free cycle with length $2^n - |F_v| - (n - 5)$ of $CQ_n - F_v$ for $|F_v| \leq 2n - 7$ and $n \geq 5$, in [5].

In this paper, we study the conditional fault-tolerance for crossed cubes. We will prove that $CQ_n - F_v - F_e$ is Hamiltonian for $|F_v| + |F_e| \leq 2n - 6$ and $n \geq 5$ if the degree of every fault-free vertex is at least 2.

The rest of this paper is organized as follows. In Section 2, we gives preliminaries about definition of crossed cube and some needed lemmas. Section 3 presents the main result. The conclusion is given in Section 4.

II. PRELIMINARIES

Definition 1 ([5]): An n -dimensional crossed cube, denoted CQ_n , is defined recursively as follows: CQ_1 is a complete graph on two nodes labelled with 0 and 1, respectively. For $n \geq 2$, let CQ_n^0 (respectively, CQ_n^1) denote an $(n - 1)$ dimensional crossed cube with the label of each node being preceded by 0 (respectively, 1). Then, CQ_n is built from CQ_n^0 and CQ_n^1 by adding all those edges (u, v) such that

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- 1) $u = u_{n-1}u_{n-2} \cdots u_0 \in V(CQ_n^0)$, $v = v_{n-1}v_{n-2} \cdots v_0 \in V(CQ_n^1)$.
- 2) $u_{n-2} = v_{n-2}$ if n is even, and
- 3) for $0 \leq i \leq \lfloor (n-1)/2 \rfloor$, $(u_{2i+1}u_{2i}, v_{2i+1}v_{2i}) \in \{(00, 00)(10, 10)(01, 11)(11, 01)\}$.

Let $u = u_{n-1}u_{n-2} \cdots u_0$, $v = v_{n-1}v_{n-2} \cdots v_0$ and (u, v) be an edge of CQ_n . We call (u, v) a d -link of CQ_n if $u_i = v_i$ for $i > d$ and $u_d \neq v_d$. For any vertex u of CQ_n , we denote the neighbor of u as $c(u)$ if $(u, c(u))$ is the $(n-1)$ -link. That is, $(u, c(u))$ is an edge of CQ_n for $u \in V(CQ_n^i)$ and $c(u) \in V(CQ_n^j)$ and $\{i, j\} = \{0, 1\}$.

The following lemma is proved in [8].

Lemma 1: Suppose that (u, v) is a d -link in CQ_n , where d is odd or $d = n-2$. Then, $(c(u), c(v))$ is also a d -link.

Let F_v be the set of faulty vertices and F_e be the set of faulty edges of CQ_n . Let $F = F_v \cup F_e$ be the set of faulty elements, and F_i is the set of faulty elements of CQ_n^i , for $i = 0, 1$. We will denote the number of faulty elements as: $f = |F|$, $f_e = |F_e|$, $f_v = |F_v|$, and f_i is the number of faulty elements in CQ_n^i , for $i = 0, 1$.

The following lemma is proved in [7].

Lemma 2: For $n \geq 3$, there exists a fault-free Hamiltonian cycle in a $CQ_n - F$ with $f \leq n-2$, and there exists a fault-free Hamiltonian path between any pair of fault-free nodes in a $CQ_n - F$ with $f \leq n-3$.

The following result is showed in [10].

Lemma 3: For any two pairs of nodes $\{x_1, x_2\}$ and $\{y_1, y_2\}$ in CQ_n , $n \geq 5$, there exist two disjoint paths P_1 and P_2 satisfying that (1) P_1 joins x_1 to y_1 and P_2 joins x_2 and y_2 , and (2) $V(P_1) \cup V(P_2) = V(CQ_n)$.

For any edge $e \in E(G)$, the graph G is *edge-Hamiltonian* if there exists a Hamiltonian cycle containing the edge e . Let F_e be the set of faulty edges of crossed edges. Let v be a vertex of the

crossed cube. Let $N(v)$ be the vertex set $\{u | (u, v) \text{ is an edge}\}$. Let $e = (a, b)$ be an edge of the crossed cube and $N(e)$ be the vertex set $\{a, b\}$. Let $S(w)$ be an arbitrary subset of $N(w)$ for $w \in (V(CQ_n) \cup E(CQ_n))$.

Lemma 4: The graph $CQ_n - F$ is edge-Hamiltonian for $F \subset (V(CQ_n) \cup E(CQ_n))$ and $|F| \leq n-2$ with $n \geq 5$.

Proof: Let $e = (a, b)$ be an arbitrary edge of $CQ_n - F$ and $f = |F|$. We will construct a Hamiltonian cycle of $CQ_n - F$ containing the edge e with $f \leq n-2$. Since the crossed cube is vertex-transitive, we can assume that e is not in CQ_n^1 . Without loss of generality, we can assume that $a \in V(CQ_n^0)$. When $f \leq n-3$, $CQ_n - F$ is Hamiltonian-connected. Thus $CQ_n - F$ is edge-Hamiltonian. We only need to prove this lemma for $f = n-2$. We will prove the lemma with induction on n . We obtain that $CQ_5 - F$ is edge-Hamiltonian for $f = 3$ with computer programs. We can assume that $CQ_k - F$ is edge-Hamiltonian for $f \leq k-2$, where $k \leq n-1$ and $n \geq 6$. Now, we consider CQ_{k+1} with $k-1$ faulty elements. Let F_0 and F_1 denote the sets of faulty elements in CQ_{k+1}^0 and CQ_{k+1}^1 , respectively. Let $E_c = \{(u, c(u)) | \text{for every } u \in V(CQ_{k+1}^0)\}$. We denote $f_i = |F_i|$ for $i = 0, 1$.

Case 1. $e \in CQ_{k+1}^0$.

Case 1.1. $f_0 = k-1$.

Let x be a faulty element of F and $F' = F - \{x\}$. Hence, $|F'| = k-2$. According to the induction hypothesis, we can obtain a Hamiltonian cycle C_0 which contains the edge e in $CQ_{k+1}^0 - F'$. Thus, there exists a fault-free Hamiltonian path $\langle u \rightarrow P_1 \rightarrow v \rangle$ in $CQ_{k+1}^0 - F$ passing through the edge e . By Lemma 2, there exists a fault-free Hamiltonian path $\langle c(v) \rightarrow P_2 \rightarrow c(u) \rangle$ in CQ_{k+1}^1 . Thus, $\langle u \rightarrow P_1 \rightarrow v, c(v) \rightarrow P_2 \rightarrow c(u), u \rangle$ forms a fault-free Hamiltonian cycle passing through the edge e of

$$CQ_{k+1} - F_v - F_e.$$

Case 1.2. $f_0 \leq k - 2, f_1 \leq k - 3$.

According to the induction hypothesis, we can obtain a Hamiltonian cycle C_0 which passes through the edge e in CQ_{k+1}^0 . Since $2^k - 2 > 2k - 6$ for $k \geq 5$, there exists an edge $(u, v) \neq e$ on C_0 such that $c(u), c(v), (u, c(u)), (v, c(v)) \notin F$. We can denote C_0 as $\langle u \rightarrow P_1 \rightarrow v, u \rangle$. By Lemma 2, there exists a Hamiltonian path $\langle c(v) \rightarrow P_2 \rightarrow c(u) \rangle$ in $CQ_{k+1}^1 - F_1$. Thus, $\langle u \rightarrow P_1 \rightarrow v, c(v) \rightarrow P_2 \rightarrow c(u), u \rangle$ forms a fault-free Hamiltonian cycle which passes through the edge e of $CQ_{k+1} - F_v - F_e$.

Case 1.3. $f_1 = k - 2$.

Let x be a vertex of CQ_{k+1}^1 where the edges incident to x are fault-free. Let $F_{c(x)} = \{(c(x), v) \mid \text{for } v \in V(CQ_{k+1}^0) \text{ and } (c(x), v) \text{ is a } d\text{-link where } d \text{ is even and } d \neq k - 1\}$. By induction hypothesis, we can obtain a Hamiltonian cycle C_0 which passes through the edge e in $CQ_{k+1}^0 - F_0 - F_{c(x)}$. We can denote C_0 as $\langle c(x), c(y) \rightarrow P_1 \rightarrow c(x) \rangle$ where $(c(x), c(y))$ is a d -link for $d = 1 \pmod{2}$ or $d = k - 1$. By Lemma 1, the edge (x, y) in CQ_{k+1}^1 is also a d -link. By induction hypothesis, we can obtain a Hamiltonian cycle $\langle y, x \rightarrow P_2 \rightarrow y \rangle$ in $CQ_{k+1}^1 - F_1$. Thus, $\langle c(x), x \rightarrow P_2 \rightarrow y, c(y) \rightarrow P_1 \rightarrow c(x) \rangle$ forms a fault-free Hamiltonian cycle which passes through the edge e of $CQ_{k+1} - F_v - F_e$.

Case 1.4. $f_1 = k - 1$.

Let z be a faulty element of F where $\{c(a), (b)\} \not\subseteq N(z)$ and $F' = F - \{z\}$.

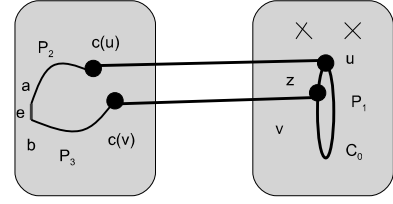


Fig. 1. The case 1.4 for edge-Hamiltonian of Lemma 4.

By induction hypothesis, there exists a Hamiltonian cycle C_1 of $CQ_{k+1}^1 - F'$. Suppose that z is on C_1 . We can denote C_1 as $\langle v, z, u \rightarrow P_1 \rightarrow v \rangle$. Suppose that z is not on C_1 . We can assume that (u, v) be an edge of C_1 . Hence, there exists a Hamiltonian path $\langle u \rightarrow P_1 \rightarrow v \rangle$ in $CQ_{k+1}^1 - F$. Suppose that $\{c(u), c(v)\} \cap \{a, b\} = \emptyset$. By Lemma 3, there exist two disjoint paths $\langle a \rightarrow P_2 \rightarrow c(u) \rangle$ and $\langle c(v) \rightarrow P_3 \rightarrow b \rangle$ of CQ_{k+1}^0 . Hence, $\langle u \rightarrow P_1 \rightarrow v, c(v) \rightarrow P_3 \rightarrow b, a \rightarrow P_2 \rightarrow c(u), u \rangle$ forms a fault-free Hamiltonian cycle passing through the edge e of $CQ_{k+1} - F_v - F_e$, as illustrated in Fig. 1.

Suppose that $|\{c(u), c(v)\} \cap \{a, b\}| = 1$. Without loss of generality, we can assume that $c(u) = a$. By Lemma 2, there exists a Hamiltonian path $\langle c(v) \rightarrow P_4 \rightarrow b \rangle$ of $CQ_{k+1}^0 - \{a\}$. Hence, $\langle u \rightarrow P_1 \rightarrow v, c(v) \rightarrow P_4 \rightarrow b, a, u \rangle$ forms a fault-free Hamiltonian cycle passing through the edge e of $CQ_{k+1} - F_v - F_e$.

Case 2. $e \in E_c$.

Without loss of generality, we can assume that $f_0 \geq f_1$.

Case 2.1. $f_0 = k - 1$.

Let z be an element of F and $F' = F - \{z\}$. By induction hypothesis, we can obtain a Hamiltonian cycle in $CQ_{k+1}^0 - F'$. Hence, there exists a Hamiltonian path $\langle v \rightarrow P_1 \rightarrow u \rangle$ in $CQ_{k+1}^1 - F_v - F_e$.

Case 2.1.1. $a \notin \{u, v\}$.

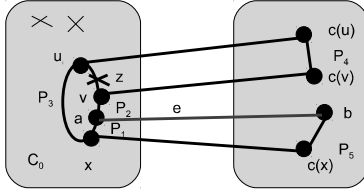


Fig. 2. The case 2.1.1 for edge-Hamiltonian of Lemma 4.

We can denote the path $\langle v \rightarrow P_1 \rightarrow u \rangle$ as $\langle v \rightarrow P_2 \rightarrow a, x \rightarrow P_3 \rightarrow u \rangle$. By Lemma 3, there exist two spanning disjoint paths $\langle c(u) \rightarrow P_4 \rightarrow c(v) \rangle$ and $\langle b \rightarrow P_5 \rightarrow c(x) \rangle$ in CQ_{k+1}^1 . Thus, $\langle u, c(u) \rightarrow P_4 \rightarrow c(v), v \rightarrow P_2 \rightarrow a, b \rightarrow P_5 \rightarrow c(x), x \rightarrow P_3 \rightarrow u \rangle$ forms a fault-free Hamiltonian cycle which passes through the edge e of $CQ_{k+1} - F_v - F_e$, as illustrated in Fig. 2.

Case 2.1.2. $a \in \{u, v\}$.

Without loss of generality, we can assume that $a = v$ and $b = c(v)$. By Lemma 2, there exists a Hamiltonian path $\langle c(u) \rightarrow P_6 \rightarrow b \rangle$ in CQ_{k+1}^1 . Thus, $\langle u, c(u) \rightarrow P_6 \rightarrow b, a \rightarrow P_1 \rightarrow u \rangle$ forms a fault-free Hamiltonian cycle which passes through the edge e of $CQ_{k+1} - F_v - F_e$.

Case 2.2. $f_0 \leq k - 2, f_1 \leq k - 3$.

Let (a, x) be an edge of CQ_{k+1}^0 such that $(x, c(x)), c(x)$ are fault-free. By induction hypothesis, we can obtain a Hamiltonian cycle $\langle x \rightarrow P_7 \rightarrow a, x \rangle$ in $CQ_{k+1}^0 - F_0$. By Lemma 2, there exists a Hamiltonian path $\langle b \rightarrow P_8 \rightarrow c(x) \rangle$ of $CQ_{k+1}^1 - F_1$. Thus, $\langle a, b \rightarrow P_8 \rightarrow c(x), x \rightarrow P_7 \rightarrow a \rangle$ forms a fault-free Hamiltonian cycle which passes through the edge e of $CQ_{k+1} - F_v - F_e$. $\square$

Lemma 5: Let v be an arbitrary vertex of CQ_n and F be a faulty set for $F \subset (V(CQ_n) \cup E(CQ_n))$. The graph $CQ_n - \{v\} - S(v) - F$ is Hamiltonian for $|F| \leq n - 4$ and Hamiltonian-connected for $|F| \leq n - 5$ and $n \geq 5$.

Proof. We obtain that $CQ_5 - \{v\} - S(v)$ is Hamiltonian-connected and $CQ_5 - \{v\} - S(v)$ is 1-Hamiltonian for any vertex $v \in V(CQ_5)$ with computer programs. We will show this lemma by induction on n . By induction hypothesis, we can assume that $CQ_k - \{v\} - S(v) - F$ is Hamiltonian for $|F| \leq k - 4$ and Hamiltonian-connected for $|F| \leq k - 5$ and $5 \leq k \leq n - 1$. We will first show that $CQ_{k+1} - \{v\} - S(v) - F$ is Hamiltonian for $|F| \leq k - 3$. Without loss of generality, we assume that v is in CQ_{k+1}^0 . We use F'_1 to denote all the set of faulty elements of CQ_n^1 . Thus, $F'_1 = (S(v) \cap V(CQ_{k+1}^1)) \cup F_1$. We will prove this result with the following cases:

Case 1. $f_0 = k - 3$

Let x be a faulty element of F and $F' = F - \{x\}$. Hence, $|F'| = k - 4$. According to the induction hypothesis, we can obtain a Hamiltonian cycle C_0 in $CQ_{k+1}^0 - \{v\} - S(v) - F'$. Thus, there exists a fault-free Hamiltonian path $\langle x_0 \rightarrow P_1 \rightarrow y_0 \rangle$ in $CQ_{k+1}^0 - \{v\} - S(v) - F$. By Lemma 2, there exists a fault-free Hamiltonian path $\langle c(y_0) \rightarrow P_2 \rightarrow c(x_0) \rangle$ in $CQ_{k+1}^1 - F'_1$. Thus, $\langle x_0 \rightarrow P_1 \rightarrow y_0, c(y_0) \rightarrow P_2 \rightarrow c(x_0), x_0 \rangle$ forms a fault-free Hamiltonian cycle of $CQ_{k+1} - \{v\} - S(v) - F$.

Case 2. $f_0 \leq k - 4$

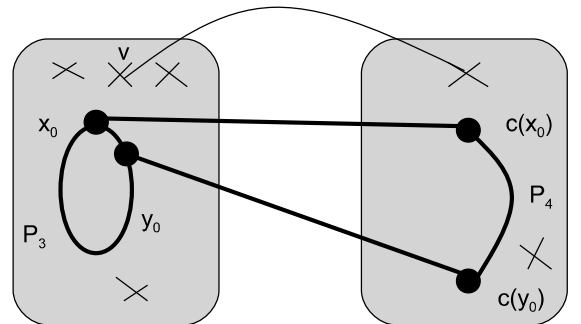


Fig. 3. The case 2 for Hamiltonian of Lemma 5.

According to the induction hypothesis, we can obtain a Hamiltonian cycle $\langle x_0 \rightarrow$

$P_3 \rightarrow y_0, x_0$ of $CQ_{k+1}^0 - \{v\} - S(v) - F_0$ for $(x_0, c(x_0)), (y_0, c(y_0)), c(x_0), c(y_0) \notin F$. By Lemma 2, there exists a fault-free Hamiltonian path $\langle c(y_0) \rightarrow P_4 \rightarrow c(x_0) \rangle$ in $CQ_{k+1}^1 - F'_1$. Thus, $\langle x_0 \rightarrow P_3 \rightarrow y_0, c(y_0) \rightarrow P_4 \rightarrow c(x_0), x_0 \rangle$ forms a fault-free Hamiltonian cycle of $CQ_{k+1} - \{v\} - S(v) - F$, as illustrated in Fig. 3.

Case 3. $f_0 \leq k - 5, f_1 \leq k - 3$

Let $(c(x_0), c(y_0))$ be an edge of CQ_{k+1}^1 for $(x_0, c(x_0)), (y_0, c(y_0)), x_0, y_0 \notin F$. By Lemma 4, there exists a fault-free Hamiltonian cycle C_0 passing through $(c(x_0), c(y_0))$ in $CQ_{k+1}^1 - F'_1$. We can denote C_0 as $\langle c(y_0) \rightarrow P_5 \rightarrow c(x_0), c(y_0) \rangle$. By induction hypothesis, we can obtain a Hamiltonian path $\langle x_0 \rightarrow P_6 \rightarrow y_0 \rangle$ of $CQ_{k+1}^0 - \{v\} - S(v) - F_0$. Thus, $\langle x_0 \rightarrow P_6 \rightarrow y_0, c(y_0) \rightarrow P_5 \rightarrow c(x_0), x_0 \rangle$ forms a fault-free Hamiltonian cycle of $CQ_{k+1} - \{v\} - S(v) - F$.

In the following, we will show that $CQ_{k+1} - \{v\} - S(v) - F$ is Hamiltonian-connected for $|F| \leq k - 4$ and $5 \leq k \leq n - 1$. Without loss of generality, we can assume that v is in CQ_{k+1}^0 . Let s and t be two different fault-free vertices of $CQ_{k+1} - \{v\} - S(v) - F$. We will construct the Hamiltonian path joining s to t of $CQ_{k+1} - \{v\} - S(v) - F$. We will also prove this result with the following cases.

Case 1. $f_0 = k - 4$.

Case 1.1. $s, t \in V(CQ_{k+1}^0)$.

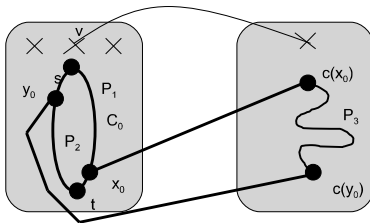


Fig. 4. The case 1.1 for Hamiltonian of Lemma 5.

According to the induction hypothesis, we

can obtain a Hamiltonian cycle C_0 in $CQ_{k+1}^0 - \{v\} - S(v) - F_0$. We can denote C_0 as $\langle s \rightarrow P_1 \rightarrow x_0, t \rightarrow P_2 \rightarrow y_0, s \rangle$. Without loss of generality, we can assume that $|P_1| \geq |P_2|$. By Lemma 2, there exists a fault-free Hamiltonian path $\langle c(x_0) \rightarrow P_3 \rightarrow c(y_0) \rangle$ of $CQ_{k+1}^1 - F'_1$. Thus, $\langle s \rightarrow P_1 \rightarrow x_0, c(x_0) \rightarrow P_3 \rightarrow c(y_0), y_0 \rightarrow P_2^{-1} \rightarrow t \rangle$ forms a fault-free Hamiltonian path of $CQ_{k+1} - \{v\} - S(v) - F$, as illustrated in Fig. 4.

Case 1.2. $s \in V(CQ_{k+1}^i), t \in V(CQ_{k+1}^j)$ for $i, j \in \{0, 1\}$ and $i \neq j$.

Without loss of generality, we can assume that $i = 0, j = 1$. According to the induction hypothesis, we can obtain a Hamiltonian cycle $\langle s \rightarrow P_4 \rightarrow x_1, s \rangle$ in $CQ_{k+1}^0 - \{v\} - S(v) - F$. By Lemma 2, there exists a fault-free Hamiltonian path $\langle c(x_1) \rightarrow P_5 \rightarrow t \rangle$ of $CQ_{k+1}^1 - F'_1$. Thus, $\langle s \rightarrow P_4 \rightarrow x_1, c(x_1) \rightarrow P_5 \rightarrow t \rangle$ forms a fault-free Hamiltonian path of $CQ_{k+1} - \{v\} - S(v) - F$.

Case 1.3. $s, t \in V(CQ_{k+1}^1)$

Case 1.3.1. $c(s)$ or $c(t)$ is fault-free.

Without loss of generality, we can assume that $c(s)$ is fault-free. By induction hypothesis, we can construct a Hamiltonian cycle $\langle c(s) \rightarrow P_6 \rightarrow x_2, c(s) \rangle$ of $CQ_{k+1}^0 - \{v\} - S(v) - F$. By Lemma 2, there exists a fault-free Hamiltonian path $\langle c(x_2) \rightarrow P_7 \rightarrow t \rangle$ of $CQ_{k+1}^1 - F'_1 - \{s\}$. Thus, $\langle s, c(s) \rightarrow P_6 \rightarrow x_2, c(x_2) \rightarrow P_7 \rightarrow t \rangle$ forms a fault-free Hamiltonian path $CQ_{k+1} - \{v\} - S(v) - F$.

Case 1.3.2. $c(s)$ and $c(t)$ are faulty.

Let $N_1(s)$ and $N_1(t)$ be the vertex sets of neighbors of s and t , respectively, of CQ_{k+1}^1 . The number of faulty vertices of CQ_{k+1}^0 is at most $|F| + |S(v)| + 1 \leq 2k - 3$. Since $c(s)$ and $c(t)$ are faulty, at most

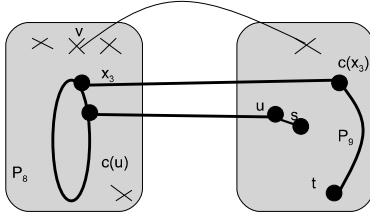


Fig. 5. The case 1.3.2 for Hamiltonian-connected of Lemma 5.

$2k - 5$ vertices of $c(N_1(s)) \cup c(N_1(t))$ are faulty. Hence, there exists a vertex $u \in (N_1(s) \cup N_1(t))$ such that $c(u)$ is fault-free. Without loss of generality, we can assume that $u \in N_1(s)$. By induction hypothesis, we can construct a Hamiltonian cycle $\langle c(u) \rightarrow P_8 \rightarrow x_3, c(u) \rangle$ of $CQ_{k+1}^0 - \{v\} - S(v) - F$. We use computer programs to verify that $CQ_5 - \{x, y, z\}$ is Hamiltonian-connected for $x, y, z \in V(CQ_5)$ and $(x, y) \in E(CQ_5)$. By Lemma 2, there exists a fault-free Hamiltonian path $\langle c(x_3) \rightarrow P_9 \rightarrow t \rangle$ of $CQ_{k+1}^1 - F'_1 - \{s, u\}$. Thus, $\langle s, u, c(u) \rightarrow P_8 \rightarrow x_3, c(x_3) \rightarrow P_9 \rightarrow t \rangle$ forms a fault-free Hamiltonian path of $CQ_{k+1} - \{v\} - S(v) - F$, as illustrated in Fig. 5.

Case 2. $f_0 \leq k - 5$ and $f_1 \leq k - 4$

Case 2.1. $s \in V(CQ_{k+1}^i), t \in V(CQ_{k+1}^j)$ for $i, j \in \{0, 1\}$ and $i \neq j$

Without loss of generality, we can assume that $i = 0, j = 1$. Let x_4 be a fault-free vertex of CQ_{k+1}^0 such that $(x_4, c(x_4))$ and $c(x_4)$ are fault-free. According to the induction hypothesis, we can obtain a Hamiltonian path $\langle s \rightarrow P_{10} \rightarrow x_4 \rangle$ in $CQ_{k+1}^0 - \{v\} - S(v) - F_0$. By Lemma 2, there exists a fault-free Hamiltonian path $\langle c(x_4) \rightarrow P_{11} \rightarrow t \rangle$ in $CQ_{k+1}^1 - F'_1$. Thus, $\langle s \rightarrow P_{10} \rightarrow x_4, c(x_4) \rightarrow P_{11} \rightarrow t \rangle$ forms a fault-free Hamiltonian path of $CQ_{k+1} - \{v\} - S(v) - F$.

Case 2.2. $s, t \in V(CQ_{k+1}^0)$

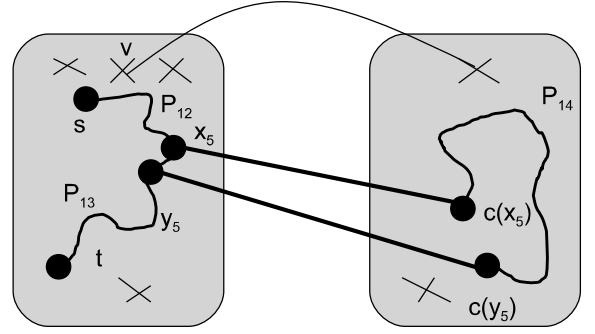


Fig. 6. The case 2.2 for Hamiltonian-connected of Lemma 5.

According to the induction hypothesis, we can obtain a Hamiltonian path between s and t of $CQ_{k+1}^0 - \{v\} - S(v) - F_0$. We can denote this path as $\langle s \rightarrow P_{12} \rightarrow x_5, y_5 \rightarrow P_{13} \rightarrow t \rangle$ such that $c(x_5), c(y_5), (x_5, c(x_5)), (y_5, c(y_5))$ are fault-free. By Lemma 2, there exists a fault-free Hamiltonian path $\langle c(x_5) \rightarrow P_{14} \rightarrow c(y_5) \rangle$ of $CQ_{k+1}^1 - F'_1$. Thus, $\langle s \rightarrow P_{12} \rightarrow x_5, c(x_5) \rightarrow P_{14} \rightarrow c(y_5), y_5 \rightarrow P_{13} \rightarrow t \rangle$ forms a fault-free Hamiltonian path of $CQ_{k+1} - \{v\} - S(v) - F$, as illustrated in Fig. 6.

Case 2.3. $s, t \in V(CQ_{k+1}^1)$

By Lemma 2, we can obtain a Hamiltonian path joining s and t of $CQ_{k+1}^1 - F'_1$. We can denote this path as $\langle s \rightarrow P_{16} \rightarrow c(x_6), c(y_6) \rightarrow P_{17} \rightarrow t \rangle$ where $x_6, y_6, (x_6, c(x_6)), (y_6, c(y_6))$ are fault-free. According to the induction hypothesis, we can obtain a Hamiltonian path $\langle x_6 \rightarrow P_{15} \rightarrow y_6 \rangle$ of $CQ_{k+1}^0 - \{v\} - S(v) - F_0$. Thus, $\langle s \rightarrow P_{16} \rightarrow c(x_6), x_6 \rightarrow P_{15} \rightarrow y_6, c(y_6) \rightarrow P_{17} \rightarrow t \rangle$ forms a fault-free Hamiltonian path of $CQ_{k+1} - \{v\} - S(v) - F$. $\square$

In fact, Lemma 5 is an important result. We can obtain that there exists a fault-free cycle with length $2^n - |F_v| - 1$ of $CQ_n - F_v$ for $F_v \subset V(CQ_n)$ and $|F_v| \leq 2n - 6$. This result improves the previous

result in [5].

III. THE MAIN RESULT

Theorem 1: Let F_e and F_v be the set of faulty vertices and faulty edges of CQ_n . The graph $CQ_n - F_e - F_v$ is Hamiltonian where the degree of every fault-free vertex is at least 2 and $|F_e| + |F_v| \leq 2n - 6$ for $n \geq 5$.

Proof. We prove this lemma with induction on n . When $n = 5$, we show the result by exhaustive search with computer programs. By induction hypothesis, $CQ_k - F_e - F_v$ is Hamiltonian where the degree of every fault-free vertex is at least 2 and $|F_e| + |F_v| \leq 2k - 6$. We will prove that $CQ_{k+1} - F_e - F_v$ is Hamiltonian where the degree of every fault-free vertex is at least 2 and $|F_e| + |F_v| \leq 2k - 4$ for $k \geq 5$. Let F_i be the faulty elements of CQ_{k+1}^i and $f_i = |F_i|$ for $i = 0, 1$. Without loss of generality, we can assume that $f_0 \geq f_1$. We will have four cases below:

Case 1. $f_0 = f_1 = k - 2$

By Lemma 1, there exists an edge (x_0, y_0) of $CQ_{k+1}^0 - F_0$ where $(c(x_0), c(y_0))$ is an edge of $CQ_{k+1}^1 - F_1$ and $c(x_0), c(y_0)$ are fault-free. By Lemma 4, there exists a Hamiltonian cycle $\langle x_0, y_0 \rightarrow P_1 \rightarrow x_0 \rangle$ in $CQ_{k+1}^0 - F_0$ and a Hamiltonian cycle $\langle c(x_0) \rightarrow P_2 \rightarrow c(y_0), c(x_0) \rangle$ in $CQ_{k+1}^1 - F_1$. Thus, $\langle x_0, c(x_0) \rightarrow P_2 \rightarrow c(y_0), y_0 \rightarrow P_1 \rightarrow x_0 \rangle$ forms a fault-free Hamiltonian cycle of $CQ_{k+1} - F_e - F_v$.

Case 2. $k - 1 \leq f_0 \leq 2k - 6, f_1 \leq k - 3$

Case 2.1. The degree of every fault-free vertex of CQ_{k+1}^0 is at least two.

According to the induction hypothesis, there exists a Hamiltonian cycle $\langle x_1, y_1 \rightarrow P_3 \rightarrow x_1 \rangle$ of $CQ_{k+1}^0 - F_0$ where $c(x_1), c(y_1), (x_1, c(x_1)), (y_1, c(y_1))$ are fault-free. By Lemma 2, there exists a Hamiltonian path $\langle c(x_1) \rightarrow c(y_1) \rangle$ of $CQ_{k+1}^1 - F_1$. Thus, $\langle x_1, c(x_1) \rightarrow P_4 \rightarrow c(y_1), y_1 \rightarrow P_3 \rightarrow x_1 \rangle$

forms a Hamiltonian cycle of $CQ_{k+1} - F_e - F_v$.

Case 2.2. There exists a fault-free vertex v of CQ_{k+1}^0 such that the degree of v in CQ_{k+1}^0 is 1.

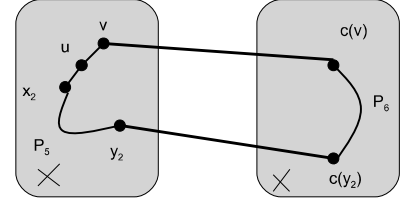


Fig. 7. The case 2.2 for Hamiltonian of Theorem 1.

Since the degree of every fault-free vertex of CQ_{k+1} is at least 2, $c(v)$ and $(v, c(v))$ are fault-free. Let u be the only fault-free neighbor of v in CQ_{k+1}^0 such that (u, v) is also fault-free. Let x_2, y_2 be fault-free vertices of CQ_{k+1}^0 such that x_2 is a neighbor of u and $c(x_2), c(y_2), (x_2, c(x_2)), (y_2, c(y_2)), (x_2, u)$ are fault-free. Let F'_0 be the set of faulty elements which is not adjacent to v . Since there are at least $k - 1$ faulty elements adjacent to v , $|F'_0| \leq k - 7$. Let $S(v) = \{z | (z, v) \in E(CQ_{k+1}^0) \text{ and } z \in F_0\}$. By Lemma 5, there exists a Hamiltonian path $\langle x_2 \rightarrow P_5 \rightarrow y_2 \rangle$ of $CQ_{k+1}^0 - \{v\} - S(v) - F'_0$. By Lemma 2, there exists a Hamiltonian path $\langle c(y_2) \rightarrow P_6 \rightarrow c(v) \rangle$ of $CQ_{k+1}^1 - F_1$. Thus, $\langle x_2 \rightarrow P_5 \rightarrow y_2, c(y_2) \rightarrow P_6 \rightarrow c(v), v, u, x_2 \rangle$ forms a Hamiltonian cycle of $CQ_{k+1} - F_v - F_e$, as illustrated in Fig. 7.

Case 3. $f_0 = 2k - 5, f_1 \leq 1$

Let x be a faulty element of F_0 and $F'_0 = F_0 - \{x\}$ such that the degree of every vertex of $CQ_{k+1}^0 - F'_0$ is at least 2. By induction hypothesis, there exists a Hamiltonian cycle $\langle x_3, x, y_3 \rightarrow P_7 \rightarrow w, x_3 \rangle$ of $CQ_{k+1}^0 - F'_0$.

Case 3.1. $\{c(x_3), c(y_3), (x_3, c(x_3)), (y_3, c(y_3))\} \subset F$.

By Lemma 2, there exists a Hamiltonian path $\langle c(x_3) \rightarrow P_8 \rightarrow c(y_3) \rangle$ of $CQ_{k+1}^1 - F_1$. Thus, $\langle x_3, c(x_3) \rightarrow P_8 \rightarrow c(y_3), y_3 \rightarrow P_7 \rightarrow w, x_3 \rangle$ forms a fault-free Hamiltonian cycle of $CQ_{k+1} - F_e - F_v$.

Case 3.2. $\{c(x_3), c(y_3), (x_3, c(x_3)), (y_3, c(y_3))\} \not\subset F$.

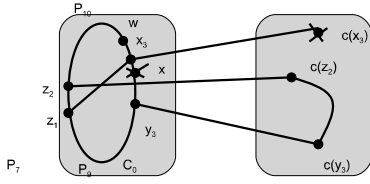


Fig. 8. The case 3.2 for Hamiltonian of Theorem 1.

Without loss of generality, we can assume that $c(x_3)$ is faulty. Since the degree of every fault-free vertex is at least 2, there are at least two fault-free neighbors of x_2 in CQ_{k+1}^0 . Let z_1 be another neighbor of x_3 such that z_1 and (x_3, z_1) are fault-free. We can denote P_7 as $\langle y_3 \rightarrow P_9 \rightarrow z_1, z_2 \rightarrow P_{10} \rightarrow w \rangle$. By Lemma 2, there exists a Hamiltonian path $\langle c(z_2) \rightarrow P_{11} \rightarrow c(y_3) \rangle$ of $CQ_{k+1}^1 - F_1$. Thus, $\langle y_3 \rightarrow P_9 \rightarrow z_1, x_3, w \rightarrow P_{10} \rightarrow z_2, c(z_2) \rightarrow P_{11} \rightarrow c(y_3) \rangle$ forms a fault-free Hamiltonian cycle of $CQ_{k+1} - F_e - F_v$, as illustrated in Fig. 8.

Case 4. $f_0 \leq 2k - 4, f_1 = 0$

Let x and y be faulty elements of F such that the degree of every fault-free vertex is at least 2. Let $F' = F - \{x, y\}$. Hence, $|F'| = 2k - 6$. According to the induction hypothesis, we can obtain a Hamiltonian cycle C_0 of $CQ_{k+1}^0 - F'$. We can denote C_0 as $\langle x_4, x \rightarrow P_{12} \rightarrow y, y_4 \rightarrow P_{13} \rightarrow x_4 \rangle$.

Case 4.1. $|P_{12}| = 1$

By Lemma 2, there exists a fault-free Hamiltonian path $\langle c(x_4) \rightarrow P_{14} \rightarrow c(y_4) \rangle$

in CQ_{k+1}^1 . Thus, $\langle x_4, c(x_4) \rightarrow P_{14} \rightarrow c(y_4), y_4 \rightarrow P_{13} \rightarrow x_4 \rangle$ forms a fault-free Hamiltonian cycle of $CQ_{k+1} - F_v - F_e$.

Case 4.2. $|P_{12}| = 2$

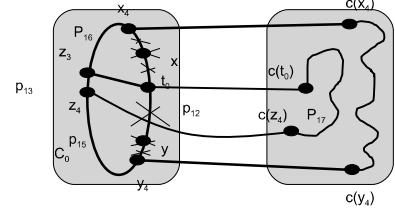


Fig. 9. The case 4.2 for Hamiltonian of Theorem 1.

We can denote the P_{12} as $\langle x, t_0, y \rangle$. Since the degree of every fault-free vertex is at least 2, there exists a neighbor z_3 of t_0 in CQ_{k+1}^0 such that z_3 and (z_3, t_0) are fault-free. We can denote P_{13} as $\langle y_4 \rightarrow P_{15} \rightarrow z_4, z_3 \rightarrow P_{16} \rightarrow x_4 \rangle$. By Lemma 3, there exist two spanning disjoint paths $\langle c(z_4) \rightarrow P_{17} \rightarrow c(t_0) \rangle$ and $\langle c(x_4) \rightarrow P_{18} \rightarrow c(y_4) \rangle$ of CQ_{k+1}^1 . Thus, $\langle t_0, z_3 \rightarrow P_{16} \rightarrow x_4, c(x_4) \rightarrow P_{18} \rightarrow c(y_4), y_4 \rightarrow P_{15} \rightarrow z_4, c(z_4) \rightarrow P_{17} \rightarrow c(t_0), t_0 \rangle$ forms a fault-free Hamiltonian cycle of $CQ_{k+1} - F_v - F_e$, as illustrated in Fig. 9.

Case 4.3. $|P_{12}| \geq 3$

We can denote the P_{12} as $\langle x, x_5 \rightarrow P_{19} \rightarrow y_5, y \rangle$. By Lemma 3, there exist two spanning disjoint paths $\langle c(x_4) \rightarrow P_{20} \rightarrow c(x_5) \rangle$ and $\langle c(y_5) \rightarrow P_{21} \rightarrow c(y_4) \rangle$ of CQ_{k+1}^1 . Thus, $\langle x_4, c(x_4) \rightarrow P_{20} \rightarrow c(x_5), x_5 \rightarrow P_{19} \rightarrow y_5, c(y_5) \rightarrow P_{21} \rightarrow c(y_4), y_4 \rightarrow P_{13} \rightarrow x_4 \rangle$ forms a fault-free Hamiltonian cycle of $CQ_{k+1} - F_v - F_e$. $\square$

IV. CONCLUSION

In this paper, we have proven that the graph $CQ_n - F_v - F_e$ is Hamiltonian where the degree of every fault-free vertex is at least 2 for $F_v \subset V(CQ_n)$, $F_e \subset E(CQ_n)$, $|F_v| + |F_e| \leq 2n - 6$ and $n \geq 5$. This result partially improves the previous

result in [8] since we prove the result under hybrid vertex and edge faults for conditional fault-tolerance Hamiltonian of CQ_n . We will investigate more conditional vertex fault-tolerance for crossed cubes, such as pancyclicity and Hamiltonian-connectivity.

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